

Fig. 2: Actual-size PCB pattern of the battery charger-cum-variable power supply

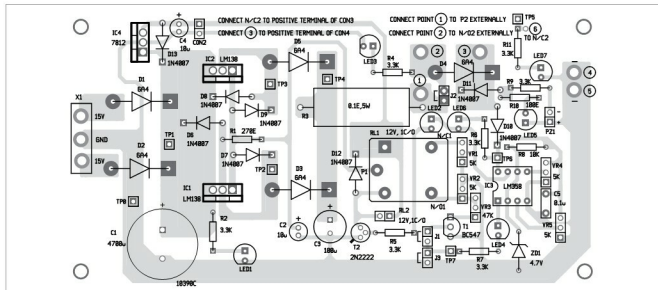


Fig. 3: Component layout of the PCB

This circuit simplifies higher current handling of LM138 by using IC1 and IC2 in parallel, but still output voltage can be regulated by a single variable resistance. Care should be taken for handling the lines carrying 5A current. Two separate relays (RL1 and RL2) are used to reduce the cost of the project.

Transformer X1 steps down 230V AC to 15V-0-15V AC that is then rectified by diodes D1 and D2 and smoothed by capacitor C1. This voltage, which is around 20V DC,

is fed to IC1 and IC2 that are wired in parallel. Their output voltage is regulated either by VR1 (in case of variable power supply) or VR2 (in case of battery charger), which is selected by relay RL2. Output so obtained is available for the variable power supply or battery charger through RL1.

Dual op-amp LM358 (IC3) is used to control the relays and select the type of output, that is, whether for battery charger or variable power supply. When no battery is connect-

ed for charging, no power is given to IC3. RL1 and RL2 are in non-energised state, and potmeter VR1 can be used to get variable voltage output across CON3 fitted on the cabinet.

When a 12V battery under charging (BUC#) is connected properly to terminal CON4 for charging, IC3 gets power supply from the battery through diode D10. If battery voltage is below 'dead' voltage (say, 6-9V and at least above 6V), pin 7 of IC3 pulls low and LED6 glows. If the battery is healthy (say, more than 9V),